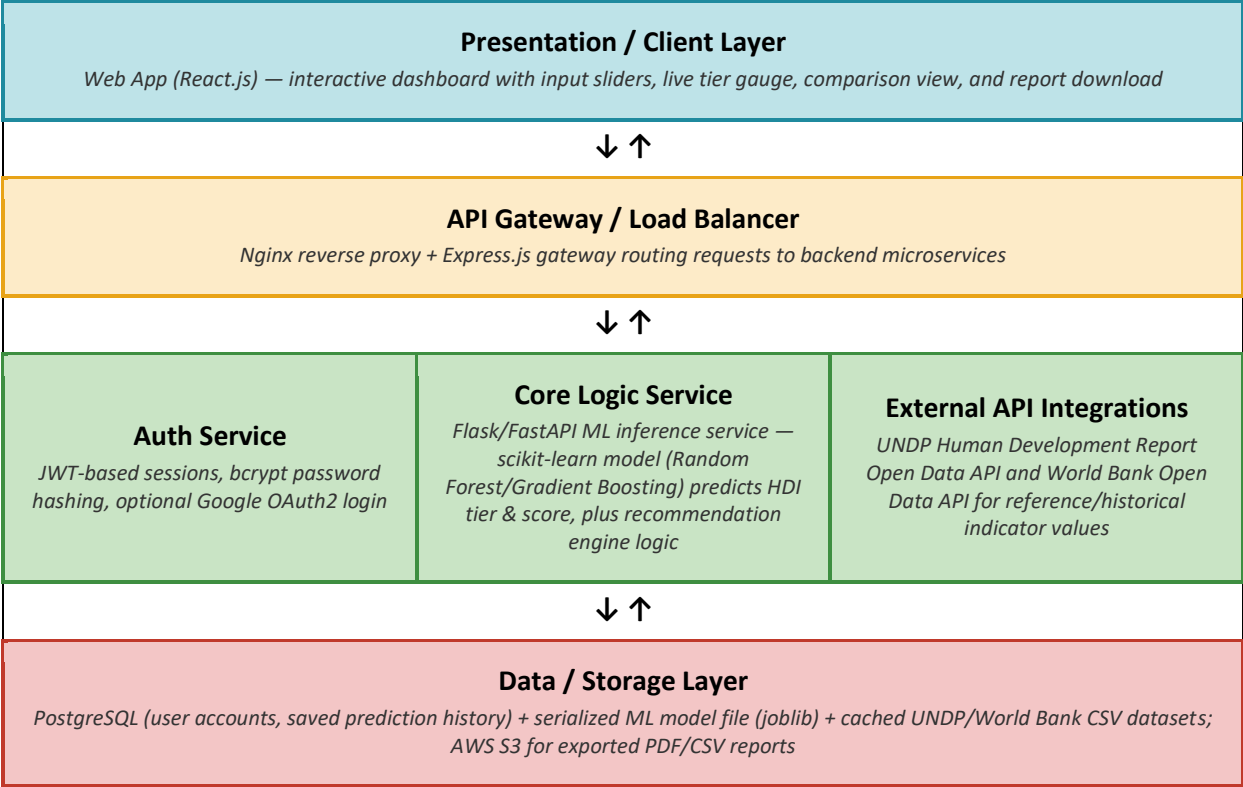


Solution Architecture

Date	08/07/ 2026
Team ID	
Project Name	HDI Predictor - Human Development Index Prediction & Recommendation Platform
Maximum Marks	5 Marks

Solution Architecture Diagram:

The HDI Predictor follows a layered architecture: a React-based presentation layer for interactive input and visualization, an API gateway routing traffic, three core backend services (Auth, ML-driven Core Logic, and External Data Integration), and a persistent data/storage layer.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders the interactive UI: 3 input fields/sliders for Life Expectancy, Education, and GNI per capita; live tier gauge, comparison view, and report download button. Calls the backend API on every input change.	React.js, Chart.js/Recharts, Axios, HTML5/CSS3
API Gateway	Single entry point for all client requests; routes traffic to Auth, Core Logic, or External Integration services, handles rate-limiting and CORS.	Nginx, Express.js
Microservices	Auth Service manages login/session; Core Logic Service loads the trained ML model, runs inference on the 3 indicators, computes the predicted tier/score, and generates the weakest-indicator	Flask/FastAPI, scikit-learn, joblib, pandas, JWT

	improvement recommendation; External Integration Service fetches reference HDI/indicator data from UNDP and World Bank APIs.	
Database	Stores registered user accounts, saved prediction history, and cached reference datasets used to retrain/validate the model.	PostgreSQL, AWS S3 (for exported reports)